



VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2015
Higher 2

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CANDIDATE NAME

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EXAM NUMBER							
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BIOLOGY

9648/02

Paper 2 Core Paper

14 September 2015

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your Class, exam number and name on all the work you hand in.
Write in dark blue or blue pen.
You may use a soft pencil for any diagrams or graphs.
Do not use any staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the question paper.

Section B

Answer any **one** question on the writing paper provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use the appropriate units.

At the end of the examination,

1. hand in sections A and B separately;
2. fasten all your work securely;
3. enter the question number of section B that you have answered in the grid opposite.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8	
Section B	
Total	

Section A: Structured questions

Answer **all** the questions. All answers must be written on the spaces provided and nowhere else.

- 1 Collagen is the main structural protein present in the various connective tissues of mammals. It is the most abundant protein in mammals.

Steps (i) to (v) in **Fig 1.1** show the main steps involved in the synthesis and assembly of collagen.

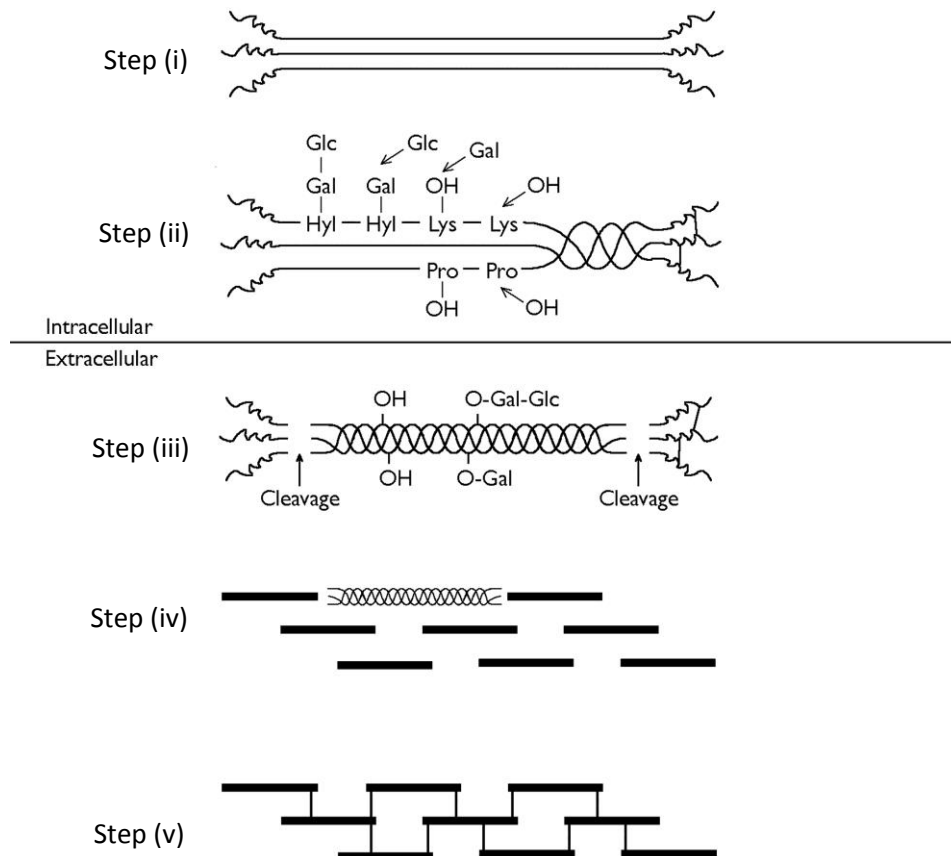


Fig 1.1

(Source: <http://iovs.arvojournals.org/article.aspx?articleid=2125662>)

- (a) Describe the primary structure of collagen.

(b) State the organelles in which steps **(i)** and **(ii)** occur in.

[2]

(c) Explain how steps **(iv)** and **(v)** contribute to the high tensile strength of collagen.

[2]

(d) Suggest why the assembly of collagen takes place outside the cell.

[2]

(e) Describe two structural differences between tropocollagen and haemoglobin.

[2]

[Total:11]

- 2 (a)** The image in **Fig. 2.1** is an electron micrograph where the boxed area is a mRNA-DNA hybrid molecule formed by mixing an antibody-coding messenger RNA (mRNA) from an immune cell with DNA that has been made single-stranded. It consists of single and double-stranded regions. The bar represents the length of 1000 bases.

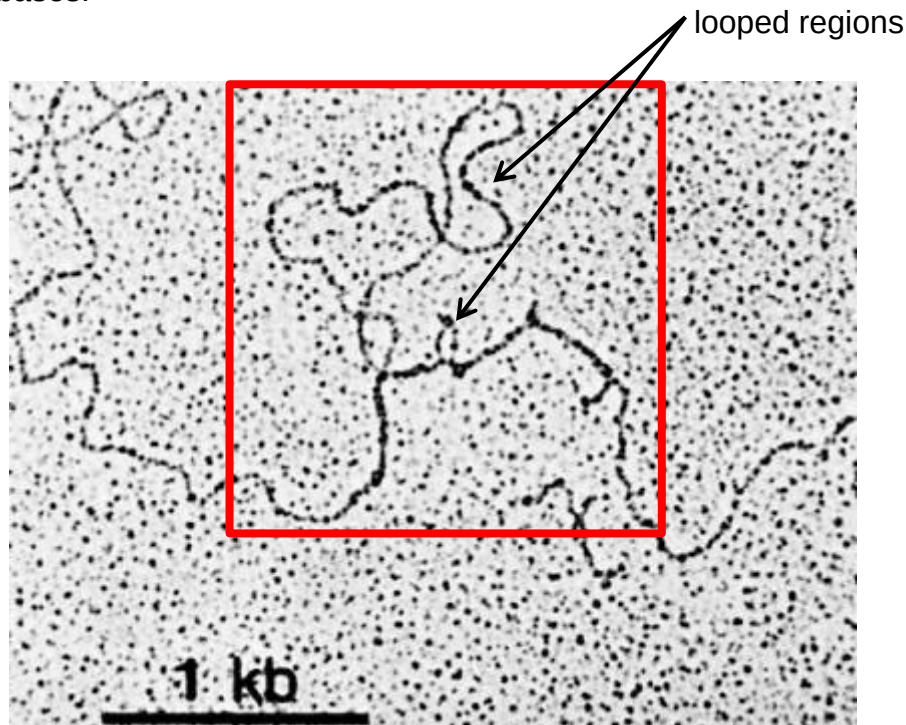


Fig. 2.1

(Source: <http://users.rcn.com/jkimball.ma.ultranet/BiologyPages/T/Transcription.html>)

- (i)** Account for the mRNA-DNA hybrid molecule in the boxed area.

- (ii) Different immune cells produce different types of antibodies. When an antibody-coding mRNA from another immune cell is mixed with the same single stranded DNA, more looped regions are formed.

Suggest why there is a difference in the mRNA-DNA hybrid molecules formed.

[2]

- (b) In order to study the process of translation, these mRNAs are introduced into different *E.coli* bacterial cells.

- (i) State one other type of nucleic acid, besides mRNA, that must be present in the bacteria in order for translation to occur. Describe clearly its role in translation.

[2]

- (ii) Describe two major differences between translation in the *E. coli* compared to that of the immune cells.

[2]

- (iii) Explain why, despite the differences, it is still possible to synthesise the protein in the *E.coli* cell.

[2]

[Total: 11]

- 3 In *E. coli*, arginine is synthesized in eight enzymatic steps. **Fig. 3.1** shows the results of an experiment carried out to determine the effects of adding arginine on the expression of some of the genes (*argA*, *argC*, *argD*, *carB*) involved in its synthesis.

The initial gene expression was measured by determining the **mRNA** produced at time zero. This is taken as 100% (white bars). All the values are relative to this initial value taken every 30 seconds over the next 2 minutes.

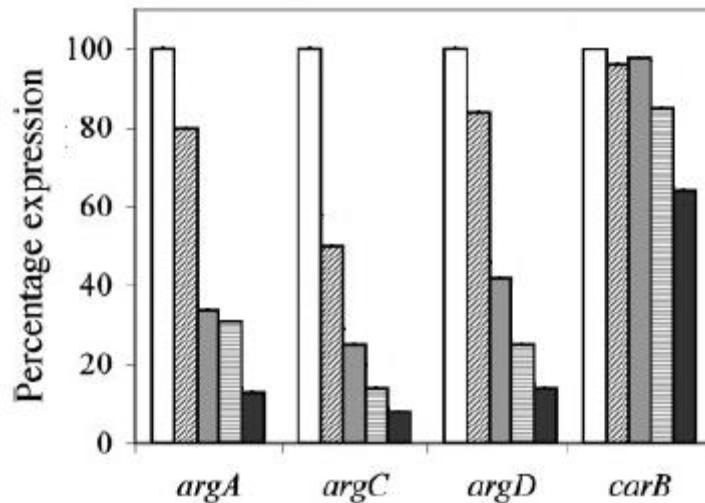


Fig. 3.1

(Adapted from <http://mic.sgmjournals.org/content/journal/micro/10.1099/mic.0.29088-0?crawler=true&mimetype=application/pdf>)

- (a) Using data from *argA*, *argC* and *argD* in **Fig. 3.1**, explain how arginine is able to control the expression of these three genes.

(b) It has been suggested that some of the genes involved in arginine synthesis are not organized into one single operon. Instead, they are scattered over different regions of the bacterial chromosome.

(i) Use evidence from **Fig. 3.1** to support the statement above.

[3]

(ii) Suggest how expression of genes not in the same operon can be regulated by arginine.

[2]

[Total: 9]

- 4 (a) An investigation was conducted on a eukaryotic chromosome to study the functions of various types of specialized DNA sequences. Two particular regions, **Q** and **V**, are shown below in **Fig. 4.1**.

Fig. 4.1 is not drawn to scale and the organisation has been simplified.

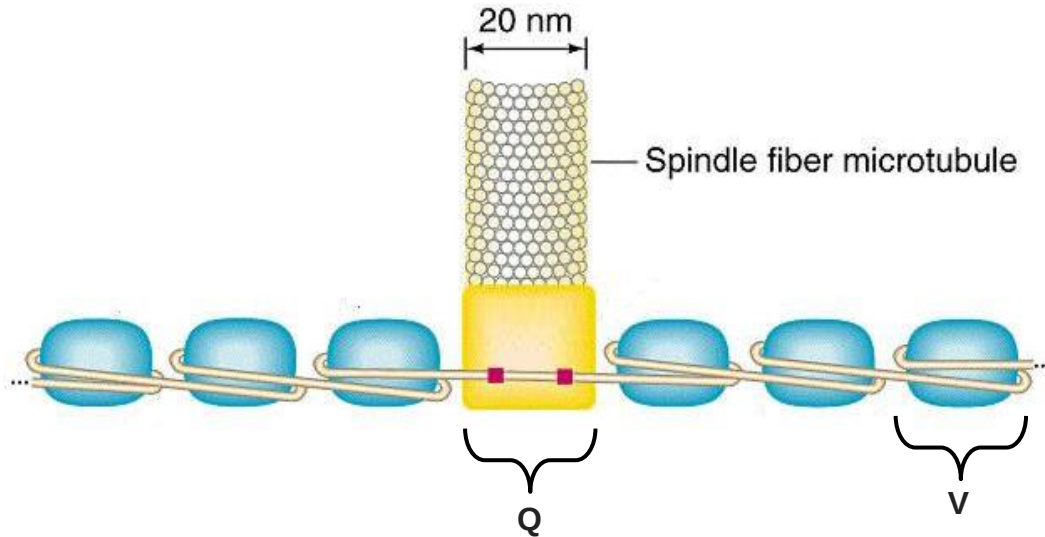


Fig. 4.1

(Adapted from <http://www.cas.miamioh.edu/~wilsonkg/old/gene2005/images/fig16p16.jpg>)

- (i) Describe how the DNA sequences found in region **Q** may differ from region **V**.

[1]

- (ii) Describe the packing of the DNA at region **Q** at the stage shown in **Fig. 4.1**.

[3]

- (iii)** A mutant organism with region **Q** deleted was shown to have reduced viability (e.g. lack important functions essential for survival). With reference to the functions of region **Q**, suggest an explanation for this observation.

[3]

- (b)** At another region on the same chromosome, a type of specialized DNA sequences was shown to be protected from exonuclease activity by a process known as 'heterochromatization'.

- (i)** Explain the significance of protecting these sequences from exonuclease activity.

[2]

- (ii)** Suggest another way in which these sequences may be protected from exonuclease activity.

[1]

[Total: 10]

- 5** Specific inherited mutations *BRCA1* and *BRCA2* increase the risk of female breast and ovarian cancers. Both *BRCA1* and *BRCA2* have DNA repair functions. Females with breast and ovarian cancers associated with *BRCA1* and *BRCA2* mutations tend to develop at younger ages than their nonhereditary counterparts.

(a) Explain the significance of mutated *BRCA1* and *BRCA2* genes in the development of cancer.

[3]

(b) Explain why individuals who inherit the *BRCA1* and *BRCA2* mutations tend to develop breast cancer at younger ages.

[3]

Blood samples collected from 5318 Jewish subjects identified the two most common mutations in *BRCA1* gene are either deletion or insertion of bases.

(c) Describe how two different mutations can bring about the same effect.

[2]

[Total: 8]

- 6** In pigeons, the males are the homogametic sex (ZZ) while the female are the heterogametic sex (ZW). Two genes are found on the Z chromosome. One gene codes for the colour of the feathers, which can be either brown or red. The other gene codes for intensity of the colouration, normal or lighter shade.

A brown and normal shade male is crossed with a red and lighter shade female and the resulting offspring are all brown and normal shade.

- (a) (i)** With the information provided, determine which allele is dominant for the colour of the feathers and the intensity of the colouration. Explain how you derived your answer.

[3]

- (ii)** Using appropriate symbols for the alleles, state the genotypes of the offspring in the above cross.

[1]

- (b)** A subsequent cross that involves a male from F1 generation and a red and lighter shade female was made. Their offspring fall into more than four phenotypic classes. Use a genetic diagram to explain the results of the cross.

- (c) To study the mode of inheritance of specific characteristics, controlled crosses are often carried out and the data of the offspring analysed using a chi-square test.

Explain the purpose of carrying out a chi-squared test and the conclusions that can be drawn from such a test.

[3]

[Total: 12]

- 7 The electronmicrograph in **Fig. 7.1** shows a junction between two neurones in the brain.

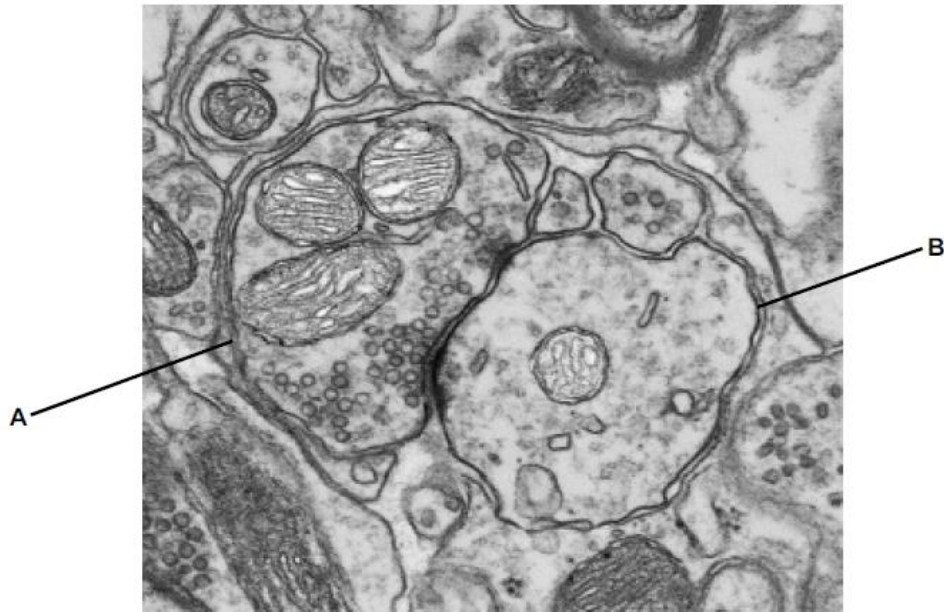


Fig. 7.1

- (a) (i) Identify neurones **A** and **B**.

A

B

[2]

- (ii) Support your answer to (i) with observable features shown in **Fig. 7.1**.
Relate these features to their functions in the transmission of nerve impulses.

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[3]

Fig. 7.2 shows the effect of the neurotransmitter dopamine binding to a post-synaptic membrane.

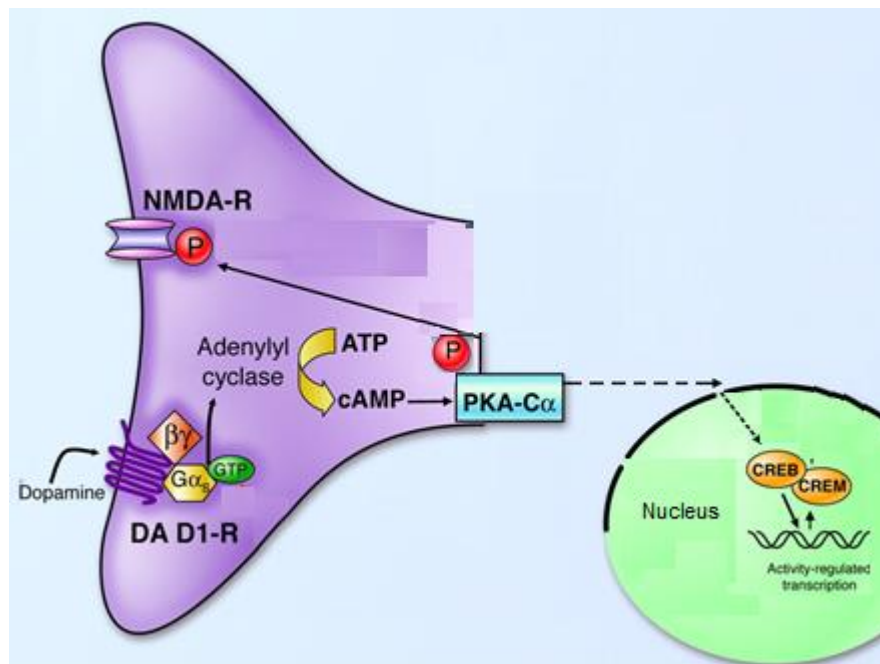


Fig. 7.2

(Adapted from *Nature Reviews Drug Discovery* **8**, 982-1001 (December 2009))

(b) With reference to **Fig. 7.2**, outline the main stages of signal transduction illustrated.

[5]

- 8 Collared lizards, *Crotaphytus collaris*, show variation in running speed and body length. **Fig. 8.1** shows a collared lizard.



Fig. 8.1

A scientist carried out an investigation on a population of collared lizards to find out whether running speed and body length affected the number of offspring that a male lizard sired (fathered). The lizards reproduce sexually, and females lay eggs after mating with males.

A large number of male lizards were captured just before mating during the breeding season.

For each lizard:

- it was chased down a small race track and its fastest running speed over 1 m recorded
- body length was measured
- a blood sample was taken and DNA sequences at 10 different loci were analysed.

The lizards were then released back into the place where they were captured for mating to commence.

Several weeks later, after one batch of the lizards' eggs had hatched, as many young hatchlings as possible were captured. DNA testing was carried out on each hatchling to determine the identity of the male parent. The results were used to determine the number of offspring sired by each of the male lizards in the first sample.

This study lasted several years with repeated sampling of the population.

(a) Explain the basis of tracking the lizard population to conclude if evolution has occurred.

[4]

(b) **Fig. 8.2** shows from the first batch of eggs, the number of offspring sired plotted against the fastest running speed (sprint speed) of the adult male lizards.

Fig. 8.3 shows from the first batch of eggs, the number of offspring sired plotted against the body length of the adult male lizards.

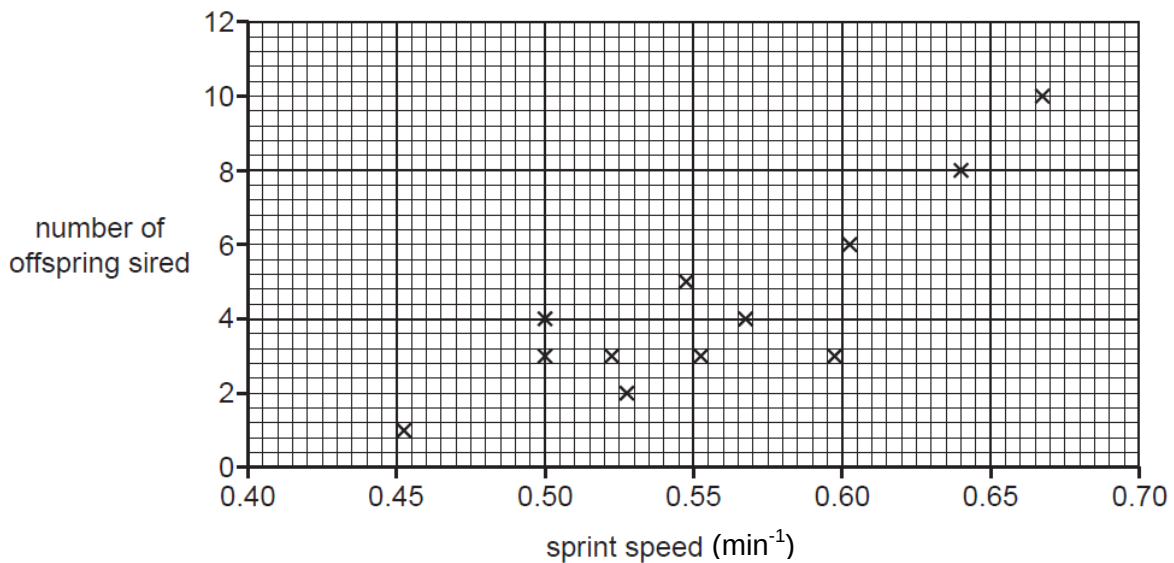


Fig. 8.2

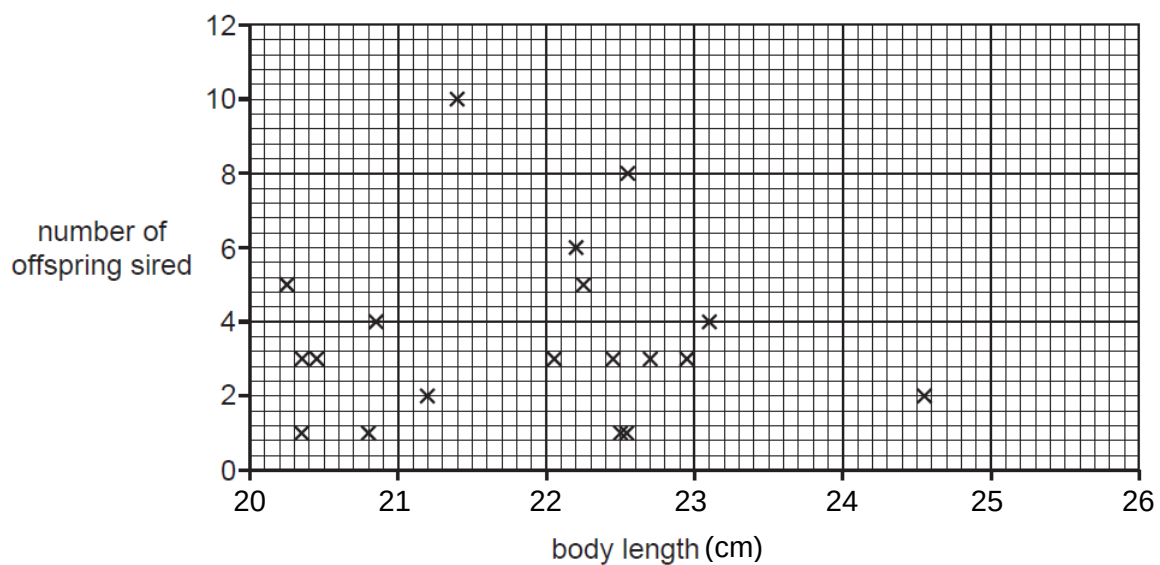


Fig. 8.3

- (i) With reference to **Fig. 8.2**, describe the relationships between sprint speed and the number of offspring produced.

- (ii) Discuss whether the data shown in **Fig. 8.3** supports the hypothesis that the body length is likely to reduce over time in the population.

[3]

[Total: 9]

Section B

Answer **one** question only.

Write your answers on separate writing paper.

Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answer must in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

Begin your answers to each part (a), (b) and (c) on a new sheet of writing papers.

- 9 (a)** Outline the roles of membranes in the regulation of blood glucose level. [8]
- (b)** State the various processes in which the concentration of active enzymes can be optimised in the cytoplasm of a eukaryotic cell. [6]
- (c)** Explain the mechanisms that genetic variation may be preserved in a natural population. [6]
- 10 (a)** Explain the mode of action of aminoacyl tRNA synthetase. [8]
- (b)** Compare and contrast photophosphorylation with oxidative phosphorylation. [8]
- (c)** When the human genome was sequenced, it was found to contain a surprising high proportion of viral sequences. Suggest how this could have happened. [4]